

MODAL OPTIMIZATION OF SMALL ROOMS USING FEM AND GENETIC ALGORITHMS

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Abstract – *This paper presents a computational tool for low-frequency modal analysis and optimization of small rooms. The method combines direct frequency-domain FEM simulation with a figure of merit based on spectral and spatial sound-field deviation. Starting from a reference geometry, it evaluates source-to-receiver responses in an audience area, selects admissible receiver combinations, and uses a genetic algorithm to explore room geometry modifications. The solver was validated semi-quantitatively against the BRAS CR2 reference case by comparing measured and simulated responses from 20 to 200 Hz. Internal tests showed that first-order FEM largely preserves the acoustic ranking obtained with second-order FEM. Compared with an exhaustive search, the genetic algorithm found low-MSFD configurations at substantially lower computational cost. The tool thus supports comparisons of acoustic configurations and informs small-room design decisions.*

Keywords – Small rooms, low frequencies, direct FEM, MSFD, genetic algorithms.

1 INTRODUCTION

In small rooms, the low-frequency acoustic response is strongly influenced by the modal behavior of the enclosed air volume. Because the wavelengths are comparable to the physical dimensions of the room, pronounced peaks, cancellations, and spatial variations may occur even between nearby listening positions. These effects are relevant in mixing rooms, recording studios, rehearsal rooms, classrooms, and other small spaces [1].

Traditionally, the modal design of small rooms has relied on dimension ratios and analytical expressions for rectangular rooms. Although these approaches provide a useful starting point, they have limitations when applied to nonrectangular geometries, displaced or inclined walls, absorbing

boundary conditions, and specific source and receiver configurations. Wave-based methods, particularly the finite element method (FEM), can solve the pressure field in geometrically complex domains [2], albeit at considerable computational cost when used in iterative optimization.

This paper presents the development of a local, open-source computational tool for low-frequency modal analysis and optimization of small rooms. The tool combines direct frequency-domain FEM simulation, evaluation using a measure of spectral and spatial sound-field deviation, automatic selection of receiver positions, and geometric exploration with a genetic algorithm. We review the conceptual background and computational method, then report semi-quantitative validation against BRAS CR2, FEM order selection, receiver-position evaluation using MSFD, and a comparison of genetic optimization with exhaustive search.

2 BACKGROUND AND CONCEPTUAL FRAMEWORK

2.1 Modal acoustics in small rooms

Small-room modal analysis has been approached through methods ranging from dimension ratios for rectangular rooms to numerical techniques for more general geometries. Bolt proposed an early approach based on the statistical distribution of normal frequencies in rectangular rooms [4]. Louden later examined favorable dimension ratios based on the distribution of natural frequencies [5], while Bonello introduced a criterion based on the progression of modal density across frequency bands [6]. These criteria remain useful starting points, although they mainly address idealized rectangular rooms.

In ideal rectangular rooms with rigid walls, modes are commonly classified as axial, tangential, or oblique according to the number of spatial axes involved in each resonance. Their natural frequencies can be approximated by

$$f_{n_x, n_y, n_z} = \frac{c}{2} \sqrt{\left(\frac{n_x}{L_x}\right)^2 + \left(\frac{n_y}{L_y}\right)^2 + \left(\frac{n_z}{L_z}\right)^2}, \quad (1)$$

where c is the speed of sound; L_x , L_y , and L_z are the room dimensions; and n_x , n_y , and n_z are nonnegative integers specifying the modal order along each axis. This expression describes the natural-frequency distribution in rectangular rooms but does not directly apply to inclined walls, breaks in the room outline, or displaced vertices. In those cases, mode shapes and natural frequencies change, and the frequency response cannot be interpreted through a simple separation of axes.

As frequency increases, the number of modes per band grows and their contributions begin to overlap. The transition from a region dominated by isolated modes to one with higher modal density is commonly estimated using the Schroeder frequency [7]:

$$f_s \approx 2000 \sqrt{\frac{T_{60}}{V}}, \quad (2)$$

where f_s is expressed in Hz, T_{60} is the reverberation time in seconds, and V is the room volume in cubic meters. In small rooms, this frequency may fall within a range of particular importance for sound reproduction, making an explicitly wave-based analysis necessary at low frequencies.

The modal pattern alone does not determine the frequency response under a particular listening condition. The contribution of each mode depends on the spatial relationship among its pressure distribution, the source, and the receiver. Consequently, the same room can exhibit very different responses for different source-to-receiver configurations [8]. Gunawan and Aditanoyo

studied inclined walls in small rooms and found that such changes can shift axial modes and alter their distribution. However, redistributing modes alone does not guarantee a smoother frequency response, because sound pressure level also depends on room geometry and evaluation positions [9].

2.2 Numerical methods, direct FEM, and MSFD

Another line of research uses wave-based numerical methods to analyze irregular rooms. Craggs used three-dimensional finite elements to obtain natural modes and frequencies in complex enclosures [10]. Shuku and Ishihara applied FEM to acoustic fields in irregular rooms [11], while Petyt, Lea, and Koopmann developed a formulation for acoustic modes in irregular cavities [12]. Milner and Bernhard later used FEM to examine nonrectangular rooms and the influence of room shape on modal distribution [13].

Here, direct FEM refers to solving the forced acoustic problem directly in the frequency domain. At each analysis frequency, the Helmholtz equation is solved with a prescribed excitation to obtain the complex pressure field. In general, the problem is written as

$$\nabla^2 p + k^2 p = -j\omega\rho_0 q \quad \text{en } \Omega, \quad (3)$$

where p is the complex pressure field, $k = \omega/c$ is the wavenumber, ω is the angular frequency, ρ_0 is the air density, and q represents a generic volumetric source. In our implementation, excitation was prescribed as a normal velocity on an equivalent spherical surface, while the room surfaces were represented by impedance boundary conditions.

To compare acoustic configurations automatically, a figure of merit is needed for the source-to-receiver responses calculated by FEM. We used a criterion based on Mean Sound Field Deviation (MSFD), a descriptor of sound-field uniformity over a listening region. It combines magnitude deviation, which describes spectral variation in individual responses, and spatial deviation, which describes variation across receiver positions [18].

The calculation uses a sound pressure level matrix $X_{SPL_{p,i}}$, where $p = 1, \dots, n_p$ indexes receiver positions and $i = 1, \dots, n_f$ indexes the discrete frequencies evaluated. Magnitude deviation is calculated as

$$MD = \frac{1}{n_p} \sum_{p=1}^{n_p} \sqrt{\frac{1}{n_f - 1} \sum_{i=1}^{n_f} (X_{SPL_{p,i}} - \bar{X}_{SPL_p})^2}, \quad (4)$$

where $\bar{X}_{SPL_p}$ is the frequency average of the sound pressure levels at receiver position p . Spatial deviation is calculated as

$$SD = \frac{1}{n_f} \sum_{i=1}^{n_f} \sqrt{\frac{1}{n_p - 1} \sum_{p=1}^{n_p} (X_{SPL_{p,i}} - \bar{X}_{SPL_i})^2}, \quad (5)$$

where $\bar{X}_{SPL_i}$ is the average across receiver positions at discrete frequency i . The two terms are then combined with equal weights:

$$MSFD = 0.5 \cdot MD + 0.5 \cdot SD. \quad (6)$$

2.3 Modal optimization with genetic algorithms

Several studies have connected modal analysis with design optimization. Cox optimized modal spacing in small rooms using merit functions based on natural frequencies [14]. Papadopoulos

proposed a FEM-based method for redistributing low-frequency acoustic modes through local geometric changes to walls [15]. Cox, D’Antonio, and Avis extended this approach by considering source and receiver placement to obtain smoother low-frequency responses [16]. More recently, Petrolli et al. presented the Non-cuboid Iterative Room Optimizer (NIRO), which uses genetic algorithms and wave-based simulation to optimize non-cuboid rooms [17].

Genetic algorithms can explore large design spaces with multiple variables without requiring an explicit analytical expression for the objective function. This is useful when the cost function has multiple local minima, nonlinear behavior, or a complex dependence on geometric variables evaluated through numerical simulation [19]. In this study, each chromosome represented a candidate room configuration, and its fitness was calculated from its MSFD.

3 COMPUTATIONAL METHODOLOGY

3.1 General evaluation procedure

The developed tool evaluates and optimizes low-frequency acoustic configurations in small rooms. Starting from a reference geometry, room variables such as vertex displacements, height, and wall inclination can be modified. Receiver combinations are selected within an audience area according to the adopted optimization criterion. Source positions are specified by the user and held fixed during each evaluation.

The internal procedure combines three-dimensional meshing, direct FEM simulation, source-to-receiver response evaluation, MSFD calculation, and genetic optimization. Figure 1 summarizes the evaluation procedure used to assign a comparable acoustic value to each candidate geometry.

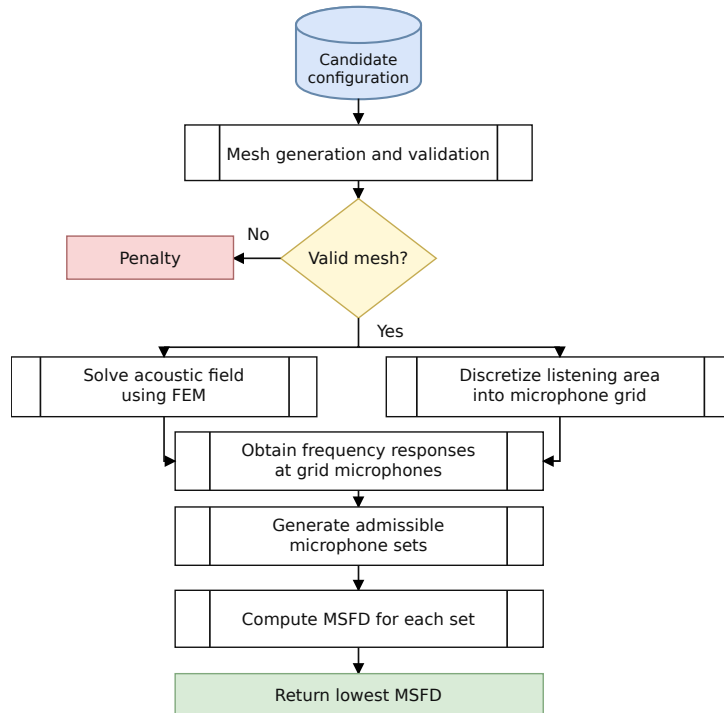


Figure 1: Evaluation procedure for a candidate configuration.

The procedure began with reconstructing the room geometry and generating a volumetric mesh. At this stage, the geometry was checked for validity and successful meshing. If it had

inconsistencies such as intersecting surfaces, an unclosed volume, or a source outside the domain, the evaluation stopped and the configuration was penalized during optimization.

At each analysis frequency, the solver calculated the complex pressure field throughout the air volume. The field was then evaluated on a regular grid of candidate receiver positions in the audience area at a fixed height of 1.20 m, representing seated listening. The complex pressures were converted to sound pressure levels in dB, and admissible receiver combinations were formed. Each combination had to satisfy the number of receivers and minimum spacing specified by the user. MSFD was calculated for each set, and the combination with the lowest value was retained.

This procedure avoided solving the FEM problem again for every receiver combination: each geometry was simulated once, and receiver combinations were then evaluated using the responses already calculated on the grid. The final output was the minimum MSFD and the receiver combination that produced it.

3.2 Meshing and direct FEM simulation

The meshing module converted the room geometry into a three-dimensional domain for FEM. Gmsh was used to construct the geometry, define physical entities, and generate the volumetric mesh [20]. The input specified a polygonal floor plan, room height, possible wall inclinations, and source position. Floor, ceiling, and wall surfaces were constructed to form a closed enclosure representing the air volume. Figure 2 shows an example of the resulting volumetric mesh.

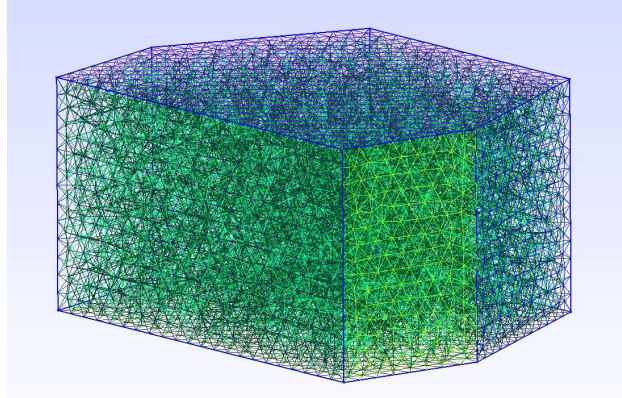


Figure 2: Example of a volumetric mesh generated from the specified room geometry.

The target element size was controlled by a characteristic length l_c linked to the shortest wavelength in the analysis range. The initial criterion was

$$h_{\max} \leq \frac{\lambda_{\min}}{6}, \quad (7)$$

where $h_{\max}$ is the maximum element size and $\lambda_{\min} = c/f_{\max}$ is the wavelength at the highest frequency evaluated. This criterion provided spatial resolution appropriate for the frequency range while balancing accuracy and computational cost. Insufficient discretization of a Helmholtz problem can introduce phase or numerical dispersion errors, especially as the wavenumber increases [21].

Using the generated mesh, the simulation module solved the acoustic problem with direct frequency-domain FEM. FEniCSx was used to construct the finite element space, assemble the variational formulation, and solve the resulting complex linear system [22]. Simulations covered 20 Hz through the maximum frequency selected by the user, with homogeneous air of density $\rho_0 = 1.225 \text{ kg/m}^3$ and sound speed $c = 343 \text{ m/s}$.

The sound source was represented by a pulsating sphere included in the geometric domain during meshing. The main evaluation procedure used a radius of 0.10 m, selected based on sensitivity analyses conducted during development. Excitation was normalized to a reference level of 94 dB SPL at 1 m in a free field. This normalization provided a common reference across simulations and did not represent calibration to a specific physical source.

The room surfaces were modeled using a global equivalent normalized impedance for walls, floor, and ceiling. The main simulations used a default real normalized surface impedance $z_n = 25$, corresponding to an approximate absorption coefficient $\alpha \simeq 0.15$. This represented a weakly absorbing boundary and avoided excessively resonant responses and conditioning problems associated with an ideal lossless room.

3.3 Genetic algorithm optimization

Geometric optimization was implemented with a genetic algorithm developed using PyGAD [23]. Each chromosome represented a candidate room configuration, with genes encoding active geometric variables such as vertex displacements, height changes, and wall inclinations. For each chromosome, the geometry was reconstructed, the acoustic evaluation procedure was run, and the lowest MSFD among admissible receiver combinations was obtained.

Because the objective was to minimize MSFD, fitness was defined as

$$F = \frac{1}{1 + \text{MSFD}}. \quad (8)$$

Thus, configurations with lower MSFD received higher fitness. The implementation used tournament selection, two-point crossover, random mutation within the admissible ranges, and elitism to retain the best solution across generations. The process ended after the specified maximum number of generations or when a saturation criterion based on consecutive generations without improvement was met.

4 RESULTS

This section presents the principal results from the development and validation of the tool, organized into four parts: semi-quantitative validation of the direct FEM solver, selection of the discretization order, receiver-position evaluation with MSFD, and comparison of the genetic algorithm with an exhaustive search.

4.1 Semi-quantitative validation with BRAS CR2

To evaluate the implemented direct FEM solver, semi-quantitative validation was performed using the BRAS CR2 reference case [24]. This nonrectangular real room provides measured impulse responses for multiple source and receiver combinations. The comparison covered the 20–200 Hz band of interest.

The numerical model was based on a simplified reconstruction preserving the general floor-plan shape, height, and relative positions of sources and receivers. Two source positions and five microphone positions yielded ten source-to-receiver responses. Room losses were represented by a global normalized surface impedance applied to walls, floor, and ceiling; the main comparison used an equivalent absorption coefficient $\alpha = 0.10$.

The source was modeled as an equivalent pulsating sphere. A relative correction for the Low channel of the measured dodecahedron, normalized at 100 Hz, approximated its spectral be-

havior. The measured impulse responses were transformed by FFT after removing their mean, without spectral smoothing, and interpolated onto the FEM solver's frequency grid.

The comparison used Pearson correlation, mean absolute error (MAE), and root mean square error (RMSE). Before calculating MAE and RMSE, the overall offset between measured and simulated responses was compensated using the median difference in dB, so these metrics primarily captured residual spectral-shape error. The same compensation was applied in the plots to emphasize spectral shape rather than absolute excitation level.

The BRAS CR2 room was reconstructed using a simplified geometry preserving its general floor plan and relative source and receiver positions. Figure 3 shows the floor plan implemented in the FEM model and the positions used for comparison.

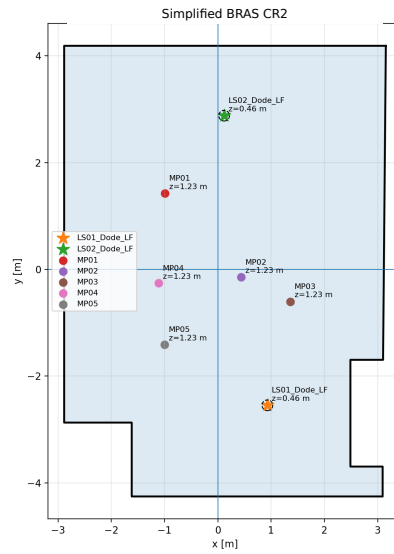


Figure 3: Simplified BRAS CR2 floor plan in the FEM model, showing source and receiver positions used for validation.

Figure 4 compares measured and simulated responses for one source position. To avoid clutter, it shows the five receivers associated with source LS01; the metrics in Table 1 were calculated from all ten source-to-receiver responses.

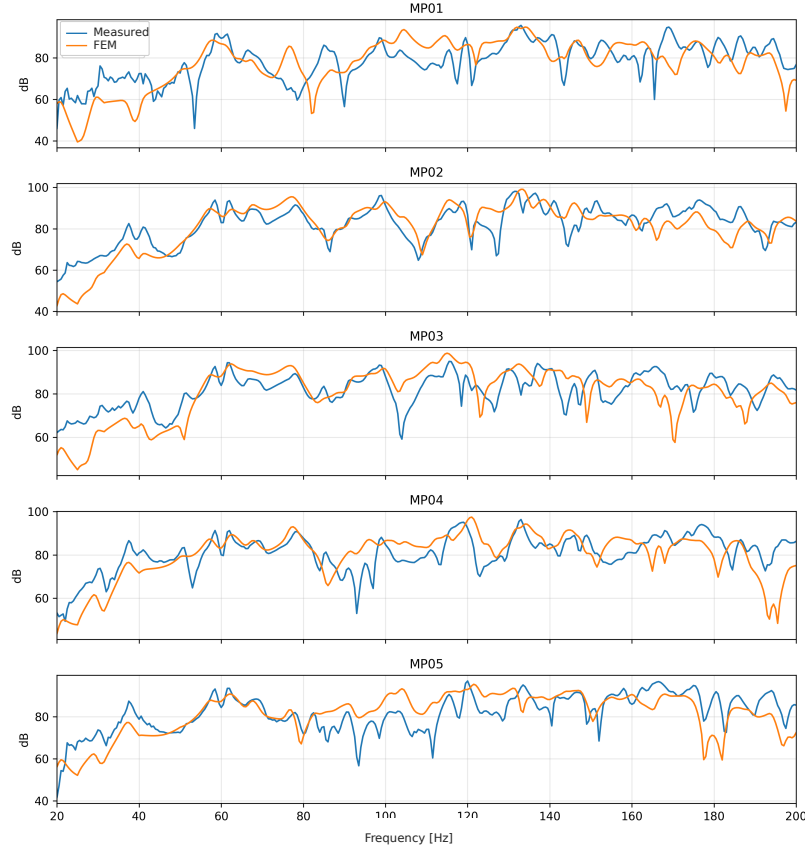


Figure 4: Measured and simulated responses for source LS01 and five microphones in BRAS CR2 over 20–200 Hz. Simulated curves include overall level compensation to facilitate comparison of spectral shape.

Table 1 summarizes the metrics obtained with first-order and second-order Lagrange elements, calculated from all ten source-to-receiver responses.

Table 1: Direct FEM solver validation over 20–200 Hz with $\alpha = 0.10$.

Order	MAE [dB]	RMSE [dB]	Correlation
P1	6.37	8.29	0.689
P2	6.32	8.22	0.700

The results showed an average correlation near 0.70, with MAE and RMSE of approximately 6 dB and 8 dB, respectively. These values were considered suitable for the scope of the work given the direct comparison with measured impulse responses, without spectral smoothing, in a mode-dominated band. The aim was to verify that the solver captured the main low-frequency trends, rather than every measured peak and dip exactly. The direct FEM solver was therefore considered semi-quantitatively validated for comparisons between configurations.

4.2 FEM order selection

After validating the direct FEM solver, the discretization order for evaluation and optimization had to be selected. The goal was not only to reproduce an individual frequency response accurately but to preserve the ranking of configurations by MSFD.

The first-order and second-order direct FEM solvers were compared over five geometries, with 200 candidate microphone positions per room and 100 four-microphone combinations per

case, reproducing the tool’s evaluation procedure. Several normalized impedances were examined through pointwise similarity between curves and correlation between configuration rankings.

Table 2: Comparison of first-order and second-order direct FEM at different normalized impedances.

z_n	Curve corr.	Pearson	Spearman	Error MSFD [dB]
2	1.00	0.99	0.98	0.35
20	0.91	0.98	0.97	0.08
40	0.89	0.98	0.98	0.10
200	0.79	0.97	0.96	0.13
400	0.75	0.94	0.94	0.17
Average	0.87	0.97	0.97	0.16

Table 2 shows that, although pointwise curve correlation decreased at higher impedances, the MSFD ranking of configurations remained highly consistent. The mean Spearman correlation was 0.97, with a mean absolute MSFD difference of 0.16 dB. The first-order direct FEM method was selected for optimization because it largely preserved the ranking obtained with the second-order method at lower computational cost. Lossless cases were not adopted as the main operating condition because the undamped model produced excessively resonant responses and conditioning problems.

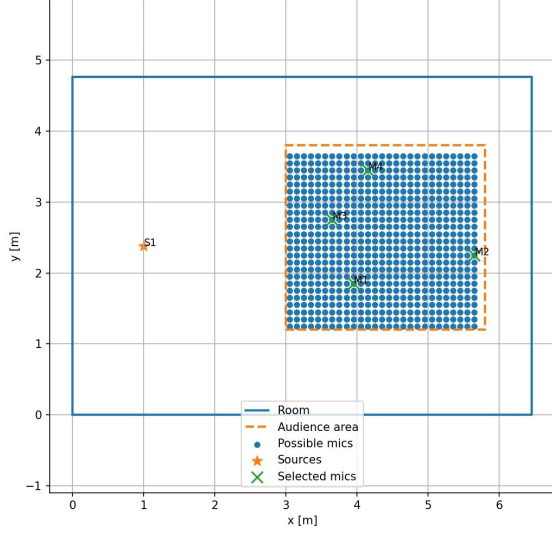
4.3 Receiver position evaluation

The next case study assessed how receiver selection affected MSFD. A reference rectangular room was defined using a conventional small-room dimension ratio consistent with geometric recommendations for critical listening. The ratio $h : w : l = 1 : 1.40 : 1.90$, associated with Loudon, also matches a favorable ratio later reported by Rindel [5, 25]. The room also meets the ITU-R BS.1116 geometric criteria for floor area and the relationships among length, width, and height [26].

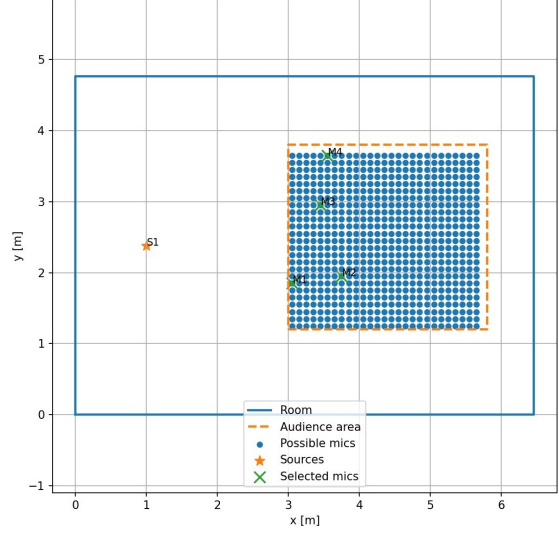
Table 3: Dimensions of the reference rectangular room used to evaluate receiver selection.

Parameter	Value
Length l	6.46 m
Width w	4.76 m
Height h	3.40 m
Ratio $h : w : l$	1 : 1.40 : 1.90
Floor area	30.75 m ²
Volume	104.55 m ³

The source was placed in the front of the room, centered across its width and at a height of 1.20 m, at $S = (1.00, 2.38, 1.20)$ m. The audience area was defined as an extended rectangle in the rear half of the room. A regular grid of candidate receiver positions was generated in this area, and admissible four-receiver combinations were evaluated with a minimum spacing of 0.70 m. Room geometry, source position, and audience area remained fixed, so the difference between evaluations arose solely from the selected receiver positions.



(a) Receiver combination with the highest MSFD.



(b) Receiver combination with the lowest MSFD.

Figure 5: Comparison of receiver combinations evaluated in the same rectangular room.

The MSFD values obtained for both receiver combinations are summarized in Table 4.

Table 4: Comparison of the highest and lowest MSFD values obtained with four receivers.

Receiver combination	MSFD [dB]
Highest MSFD	6.219
Lowest MSFD	4.496

The comparison showed that, even in a room with favorable dimensions consistent with recommendations for critical listening, the MSFD varied substantially depending on the receiver positions selected. This suggests that conventional room dimensioning criteria provide a useful starting point, but cannot, on their own, describe the acoustic performance of a specific listening arrangement. By accounting for the source-to-receiver transfer functions, the MSFD evaluation identified listening positions with lower spatial and spectral deviation within the same audience area.

4.4 Geometric optimization

To evaluate the performance of the genetic algorithm, we compared its results with those of a discretized exhaustive search over a bounded range of room geometries. This test aimed to determine whether the algorithm could find configurations with low MSFD and whether it converged toward solutions consistent with a systematic search of the same design space.

We defined a symmetric reference room measuring 5 m in width, 4 m in depth, and 3 m in height. The source was placed at (0, 3, 1.5) m, while the audience area remained fixed and centered within the room. Before optimization, this reference geometry was evaluated using the same procedure applied throughout the experiment: the mesh was generated, the direct FEM simulation was run, and the best admissible combination of four receivers in the audience area was selected, with a minimum spacing of 0.5 m. The resulting baseline MSFD was 4.74 dB.

The search space was defined by varying vertices V_2 and V_3 , with displacements Δx and Δy ranging from -0.50 m to 0.50 m. Symmetry about the room's longitudinal axis was imposed: V_1 was obtained by reflecting V_2 , and V_4 by reflecting V_3 . The height Z could also vary from 3.0 m to 4.2 m, while the walls remained vertical.

For the exhaustive search, the continuous design space was discretized in steps of 0.25 m for vertex displacements and 0.20 m for height, yielding 4.375 room geometries. Each geometry was evaluated using the same procedure: mesh generation, a first-order direct FEM simulation, selection of the best combination of four receivers with a minimum spacing of 0.5 m, and MSFD calculation over the 20–200 Hz band. The genetic algorithm explored the same parameter ranges, but its genes were not restricted to the discrete points of the exhaustive-search grid. Thus, the two approaches differed in how they explored the design space, while using the same acoustic evaluation procedure.

Table 5: Comparison of the reference room, the best exhaustive-search solution, and the best genetic-algorithm solution.

Case	Computation time	MSFD [dB]	Improvement [dB]
Reference room	–	4.74	–
Exhaustive search	16 h 46 min 11 s	2.62	2.12
Genetic algorithm	56.62 min	2.48	2.26

Both approaches substantially improved on the reference room. The genetic algorithm achieved a slightly lower final MSFD than the best solution found by the exhaustive search, by approximately 0.14 dB. The difference in computation time was more pronounced: the exhaustive search took nearly 17 hours to evaluate 4.375 geometries, whereas the genetic algorithm found a better solution in less than one hour.

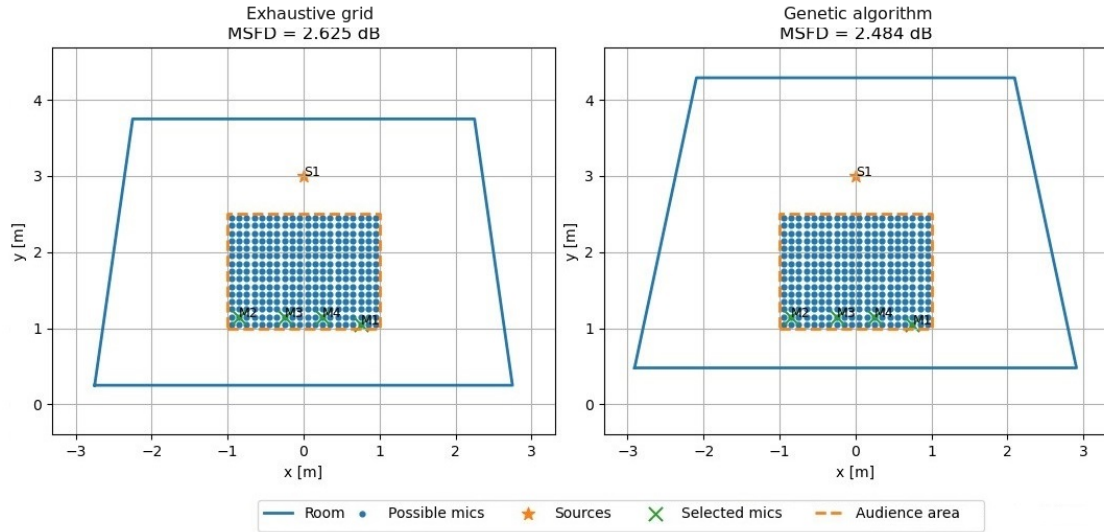


Figure 6: Comparison of the best room geometries found by the exhaustive search and the genetic algorithm.

Table 6 summarizes the vertex coordinates and heights of the two optimized solutions.

Table 6: Vertex coordinates and heights of the best solutions found by the exhaustive search and the genetic algorithm.

Parameter	Exhaustive search [m]	Genetic algorithm [m]
V_1	(−2.75, 0.25)	(−2.91, 0.48)
V_2	(2.75, 0.25)	(2.91, 0.48)
V_3	(2.25, 3.75)	(2.09, 4.29)
V_4	(−2.25, 3.75)	(−2.09, 4.29)
Height Z	3.60	3.53

Both methods converged toward a symmetric trapezoidal room geometry. Although the exact vertex coordinates differed, the resulting rooms had similar geometric features, supporting the consistency of the result. Another finding was that the exhaustive search and the genetic algorithm selected the same optimal combination of receiver positions.

Figure 7 shows how the genetic algorithm’s fitness evolved across generations. Because fitness was defined as an inverse function of MSFD, an increase in fitness corresponds to a decrease in MSFD.

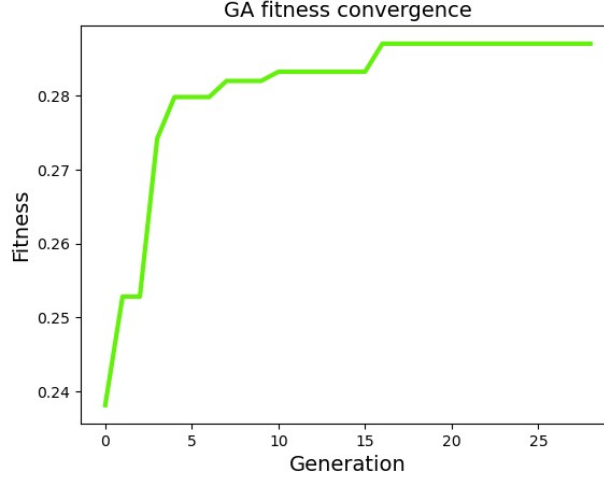


Figure 7: Convergence of the genetic algorithm’s fitness during optimization.

Most of the improvement occurred in the first few generations. Later generations progressively refined the solution until fitness stabilized. This early improvement suggests that promising room geometries could be identified well before the full, approximately one-hour run was complete.

Overall, this experiment showed that the genetic algorithm converged toward a region of low MSFD consistent with that identified by the discretized exhaustive search, while requiring substantially less computation time. By exploring the geometric design space continuously, it found a slightly better solution in considerably less time. These results support its use for room geometry optimization within the proposed procedure, particularly in larger design spaces where a finely discretized exhaustive search would be impractical.

5 CONCLUSIONS

This paper presented an open-source computational tool for low-frequency modal analysis and optimization of small rooms. The proposed approach combines direct frequency-domain FEM simulation, evaluation of source-to-receiver responses using MSFD, automatic selection of receiver positions, and room geometry optimization using a genetic algorithm. It therefore enables acoustic configurations to be compared beyond conventional dimensioning criteria for ideal rectangular rooms.

The implemented direct FEM solver was validated semi-quantitatively using the BRAS CR2 reference case. Although the model did not reproduce every peak and dip in the measured responses exactly, it captured the main spectral trends reasonably well over the 20–200 Hz band. This level of agreement is appropriate for the scope of the study: the tool was developed as a numerical basis for comparing configurations under consistent criteria, rather than for absolute prediction of a real room’s response.

The FEM order selection tests showed that the first-order direct method (P1) largely preserved the acoustic ranking obtained with the second-order method, particularly when configurations were ranked by MSFD. For this reason, the first-order method was adopted in the main evaluation and optimization procedure as a balance between numerical consistency and computational cost.

The tests also showed the importance of evaluating acoustic performance through source-to-receiver transfer functions, rather than relying solely on room geometry or global modal criteria. In the reference rectangular room, different receiver combinations within the same audience area produced substantially different MSFD values. Thus, even a room with favorable dimensions can perform differently depending on the listening positions considered.

Finally, the comparison with the exhaustive search confirmed that the genetic algorithm could find low-MSFD configurations within the bounded design space. It reached a solution consistent with the region identified by the exhaustive search, while exploring the design space continuously and requiring substantially less computation time. This supports its use for optimization problems in which a finely discretized exhaustive search would be computationally impractical.

Future work could incorporate more detailed impedance models, including complex, frequency-dependent boundary conditions assigned to individual surfaces. The optimization space could also be extended to include source position and localized acoustic treatments. Parallel processing, result caching, and adaptive mesh refinement could reduce computation time. Finally, a geometric sensitivity analysis could identify which room variables have the greatest effect on low-frequency MSFD.

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